

Pre-Requisites

To effectively understand this session, learners are expected to have:

- Basic Python programming knowledge
- Familiarity with environment variables
- Prior exposure to:
 - Basic LLM response generation using LLM APIs such as Gemini and HuggingFace Inference Client
 - Basic LLM evaluation concepts

Session Summary

This 180-minute live session is a coding-focused workshop on working with production-level LLM APIs.

The session begins with API key management and secure configuration practices. Learners will then build progressively richer examples using APIs for getting LLM responses.

The session emphasises on correct request formatting, parameters, setting up multi-turn conversations, getting structured outputs and understanding trade-offs in third-party hosted vs local models. Guided coding demonstrations and live experimentation are aimed to provide a good understanding of the concepts.

Learning Objectives/Outcomes

By the end of this session, learners will be able to:

- Create and securely manage API keys across Gemini, Hugging Face, and OpenAI
- Structure API requests correctly using SDK-based APIs for different providers
- Configure generation parameters for controlled outputs
- Compare API patterns from different providers
- Download and run a small local LLM using HuggingFace
- Apply a simple evaluation metric to compare outputs

Brief Agenda

Learning Objective	Comments	Duration	Mode
Secure API Setup	Create keys, manage environment variables, understand billing & usage	25 mins	Guided Coding
Gemini and HF APIs	Request structure, parameters, output formats, tools, modalities and other available tasks Available models, providers, structured chat	90 mins	Guided Coding
Local Model Execution	Download a small model from HF and run basic inference locally	30 mins	Guided Coding
OpenAI API (Overview)	Highlight differences from Gemini & HF	15 mins	Demonstration
Evaluation Example	Perform evaluation	10 mins	Guided Coding
Wrap-up & Q&A	Wrap up and learner questions	10 mins	Q&A

Detailed Agenda

Learning Objective	Comments	Duration	Mode
Secure API setup and configuration	• Demonstrate creation of API keys on Gemini, Hugging Face, and OpenAI platforms. • Explain differences in key scope, access levels, quotas, and billing dashboards. • Create a .env file and install python-dotenv. • Load environment variables securely in Python. • Show how to test API connectivity. • Highlight unsafe practices such as hardcoding keys and explain risks.	25 mins	Guided Coding

Implement and explore Gemini API capabilities	• Build a minimal Gemini request using model and contents. • Modify generation parameters such as temperature, top_p, and max_output_tokens and observe behavioural changes. • Show different inputs, try adding files with the prompts and so on to create more complex prompts • Implement structured JSON outputs and parse structured responses. • Inspect response metadata including candidates and usage details. • Explain system and user roles (model and user, in Gemini) • Build a multi-turn conversation manually by appending history or using the chat module • Demonstrate tool calling with a simple function example. • Provide a brief demonstration of additional modalities such as image/videos inputs and outputs; also cover other tasks related to text	50 mins	Guided Coding
Use Hugging Face Inference API for hosted models	• Introduce the InferenceClient and demonstrate model discovery and provider selection. • Execute a basic hosted chat completion request. • Show how (not just open source) third party models can be accessed via Hugging Face APIs as well. This helps avoid vendor lock-in • Recreate similar examples from the Gemini section using system and user messages. • Tune generation parameters and compare outputs. • Implement structured outputs.	40 mins	Guided Coding

	- Inspect and interpret response objects and metadata. - Explain additional available tasks such as summarisation or classification or other modalities.		
Download and run a local model from Hugging Face	- Explain conceptual differences between hosted and local inference. - Download a small model such as TinyLlama from Hugging Face. - Show how read access level tokens are used here - Also, some models might be gated and need an access request - Load the model using transformers. Run a basic inference example. - Compare latency and resource usage with hosted APIs. - Discuss trade-offs including hardware constraints, privacy benefits, lack of tool calling, and deployment considerations.	30 mins	Guided Coding
Understand OpenAI API differences (delta view)	- Implement a minimal OpenAI API example. - Highlight structural similarities with Gemini and Hugging Face implementations. - Explain differences in request format, parameters, and response handling. - Briefly demonstrate tool calling syntax differences. Emphasise cross-platform mental model alignment rather than platform-specific memorisation.	15 mins	Demonstration
Apply a simple automated evaluation metric	- Generate responses from two different models using the same prompt. - Store outputs programmatically. - Compute a simple ROUGE score on a summarisation task using an evaluation library.	10 mins	Guided Coding

	• Similarly, show BLEU (translation), BERTScore (semantic) etc. Show other important metrics as well. • Interpret the scores. • Discuss important factors while evaluating LLM responses like factuality, relevance, meaning etc. • Clarify limitations of automated metrics and why human evaluation remains important.		
Wrap up	Basic wrap up and doubt resolution	10 mins	Q&A

Instructor Notes

- Keep supporting files, prompts and codes handy to demonstrate quick examples
- Keep a track of time and API usage limits also while demonstrating with examples
- Show different ways to do different tasks other than simple chat or text generations, but keep the focus on chat/text only
- Clear the distinction and tradeoff between self-hosted, provider-hosted (closed vs open source)
- Focus on structured outputs, building conversations, function calling (non-agentic) etc.
- Give small examples with a variety of prompts, but keep the major ones consistent across the demo to reduce confusion

Conclusions

By the end of this live session, learners will be able to:

- Build real API-based LLM workflows
- Manage authentication securely
- Compare Gemini, Hugging Face, and OpenAI
- Run a local model
- Evaluate model outputs

Additional Readings

- [Gemini API Documentation](#)
- [Gemini API reference](#)
- [Google Gen AI Python SDK documentation](#)
- [HuggingFace Inference Providers - Your First Inference Provider Call](#)
- [HuggingFace - Chat Completion](#)

- [Hub Python Library - Inference Endpoints](#)
- [OpenAI API Platform Documentation](#)